NEU-PMC PTE LTD

CABLE JIG TESTER

Cable Continuity Test System

User Guide | v1.1

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1. Introduction

The Cable Jig Tester is a dedicated continuity test system designed by NEU Corporation to verify signal and GND connections of multi-pin cable assemblies. It connects via USB to a host PC running the Cable Jig Tester PC Software, providing rapid pass/fail results, detailed connection maps, and Excel test reports. The software supports multiple cable types via saved profiles, tests cables in either insertion orientation, and can capture the raw netlist of an unknown cable.

1.1 Key Features

- 116-pin connector verification (J1 ↔ J2).
- Automatic GND-bus detection and signal-pair validation.
- Orientation-independent testing — a good cable PASSES whether it is inserted A→B or reversed B→A; the detected orientation is labelled on the result banner and report.
- Cable profiles — learn and save any cable type, then test future cables against the saved reference.
- Raw capture mode — inspect an unknown cable with no reference; connections are auto-grouped by electrical net (GND groups and signal pairs).
- PASS / FAIL verdict with per-pin status.
- Excel report (template layout) with full “All Borders” tables and four sheets: Test Report, Connection Map, Test History, Pin Map.
- Firmware version reported over USB and shown on every report.
- Test history logging and auto-incrementing serial numbers.

1.2 System Overview

The system consists of two hardware components and one software application:

- Hardware Jig Board — contains the MCU (dsPIC30F4011), 8-channel MUX arrays for the J1 and J2 sides, power supply, and USB-UART bridge.
- Cable Under Test — the cable assembly inserted between the J1 and J2 pogo-pin connectors on the jig.
- PC Software — a Python/Tkinter application (cable_tester_app.py), also available as a standalone Windows executable (NEU_CABLE_JIG_TEST.exe), that communicates with the jig over serial, displays results, and generates Excel reports.

2. Hardware Overview

2.1 Jig Board Layout

The jig board features two pogo-pin connector blocks (J1 and J2), clearly labelled on the PCB silkscreen. Key components are described below:

Connector / Component	Description
J1 Connector Block	Left pogo-pin array — connects to the cable's first end. Rows A–G, 116 pins total.
J2 Connector Block	Right pogo-pin array — connects to the cable's second end. Same 116-pin layout.
12 V Power Input	Barrel connector (bottom-left). Supply a regulated 12 V DC, minimum 500 mA.
USB Cable Connector	USB Type-B port (bottom-centre). Connects to the PC for data and firmware control.
POWER LED	Illuminates green when 12 V is present and the MCU is running.

TEST / RESET Buttons	Two tactile push-buttons for triggering tests and resetting the MCU.
Status LEDs (x4)	Indicate test state: idle, running, pass, fail.

2.2 Power Requirements

Input Voltage	12 V DC ± 5 %
Input Current	≤ 500 mA (typical 200 mA)
Connector	2.1 mm barrel jack, centre positive
USB	USB 2.0 Type-B (data only — NOT used for power)

NOTE: *Always apply 12 V power BEFORE connecting the USB cable to avoid enumeration errors.*

3. Software Installation

3.1 System Requirements

Item	Requirement
Operating System	Windows 10 / 11 (64-bit)
Python	3.9 or later (only if running from source)
Python packages	pyserial, openpyxl
USB Driver	FT232 VCP driver (FTDI) — included in Windows Update
Screen resolution	1280 × 720 minimum

3.2 Installing Python Dependencies

Open a Command Prompt and run:

```
pip install pyserial openpyxl
```

3.3 Installing the USB Driver

- Connect the jig board to the PC via USB.
- Windows will automatically search Windows Update for the FT232 VCP driver.
- If not found automatically, download the latest VCP driver from ftdichip.com and run the installer.
- After installation, the device will appear as a COM port (e.g., COM3, COM7) in Device Manager.

NOTE: *To check the assigned COM port: open Device Manager → Ports (COM & LPT) → USB Serial Port (COMx).*

3.4 Running as a Standalone EXE (optional)

If you prefer not to install Python, you can build a self-contained Windows executable. In the src\PC_APP folder, double-click build_exe.bat (or run the commands below). The report template is bundled into the EXE automatically.

```
cd src\PC_APP
pip install pyinstaller pyserial openpyxl
pyinstaller NEU_CABLE_JIG_TEST.spec
```

The executable is created at: src\PC_APP\dist\NEU_CABLE_JIG_TEST.exe

4. Launching the Software

- Step 1: Ensure the jig board is powered (12 V connected and POWER LED is green).
- Step 2: Connect the USB cable from the jig board to the PC.
- Step 3: Start the software — either double-click NEU_CABLE_JIG_TEST.exe, or from source run:

```
python cable_tester_app.py
```

- Step 4: The main application window opens.
- Step 5: In the COM Port drop-down, select the port assigned to the jig board (e.g., COM7). Click Connect.
- Step 6: The status bar shows 'Connected to COMx'. The firmware version is displayed automatically.

NOTE: *If the COM port list is empty, click the Refresh button next to the drop-down.*

5. Performing a Continuity Test

5.1 Inserting the Cable

- Step 1: Raise the ZIF lever (if fitted) or remove any previously tested cable from the pogo-pin connectors.
- Step 2: Insert the cable under test so that one end sits on the J1 pogo-pin block and the other end on the J2 block.
- Step 3: Lower the ZIF lever / apply gentle downward pressure to ensure all pogo pins make contact.

NOTE: *Either cable end may go to J1 or J2 — the software detects the orientation automatically (see 5.4).*

5.2 Starting the Test

- Step 1: Select the correct cable profile in the Profile drop-down (see Section 6). For a brand-new cable, choose "— Raw capture (no reference) —" (see Section 7).
- Step 2: Enter or confirm the Serial Number in the Serial Number field.
- Step 3: Click the START TEST button (or press the TEST button on the jig board).
- Step 4: The progress bar animates while the firmware scans all 116 pins (~9 seconds).
- Step 5: When complete, the result banner shows PASS (green) or FAIL (red).

5.3 Reading the Result

Display Element	Meaning
Green PASS banner	All expected signal pairs detected correctly and the GND bus verified (in either orientation).
Red FAIL banner	One or more signal pairs missing, mis-wired, or shorted.
Connection table	Lists each pin pair: GND group (merged, yellow) first, then signal rows (white/blue).
✓ in J2 Pin column	Connection matches the expected wiring.
✗ (exp J2.Xx) in J2 col	Detected a different J2 pin than expected — possible mis-wire.
MISSING in J2 col	Expected signal pair not detected — possible open circuit.

NOTE: *The GND group row lists all J1 GND pins and their J2 counterparts, grouped one connector row (A, B, C…) per line.*

5.4 Cable Orientation (A → B and B → A)

A cable can be inserted either way round. The software compares the measured netlist against the profile in both directions and keeps whichever matches, so a correctly-made cable PASSES regardless of which end goes to J1 or J2:

- A→B (normal) — the cable's A end is on J1 and its B end on J2.
- B→A (reversed) — the ends are swapped; every J1 ↔ J2 pair is the mirror of the normal wiring.

The detected orientation is shown on the result banner and on the report's PASS/FAIL badge, for example: “Orientation: B → A (reversed)”. A genuinely faulty cable still FAILS in both orientations — orientation detection never hides a real fault.

6. Cable Profiles

A profile is the expected connection map (“golden” netlist) for one cable type. The software ships with the built-in Default cable (66 signal pairs + GND bus) and lets you add your own.

6.1 Selecting a Profile

Use the Profile drop-down at the top of the window to choose the cable type to test against. The list contains the Default cable, the Raw capture entry, and every profile you have saved.

6.2 Learning a New Cable

- Step 1: Insert a known-good sample of the new cable.
- Step 2: Select “— Raw capture (no reference) —” and run a test to capture the netlist (see Section 7).
- Step 3: Click “Learn current cable as profile” and enter a name.
- Step 4: The software auto-detects signal pairs (1-to-1 wires) and the GND bus (multi-pin nets) and saves the profile as a JSON file.
- Step 5: Select the new profile from the drop-down to test further cables of that type with full PASS/FAIL and orientation handling.

7. Raw Capture Mode

Raw capture mode shows exactly what the tester measured, with no reference and no PASS/FAIL judgment — ideal for a brand-new or unknown cable on its first run.

- Select “— Raw capture (no reference) —” in the Profile drop-down.
- Run the test. The CONNECTIONS FOUND window lists the measured netlist grouped by electrical net.
- Each 2-pin net is shown as a single signal pair (J1.x ↔ J2.y), sorted numerically.
- Each multi-pin net is shown as a GROUP header followed by its member pins, one connector row (A, B, C…) per line, J1 side and J2 side aligned side-by-side.
- Multiple independent groups are detected and listed separately.
- Click “Learn current cable as profile” to save the captured netlist for future PASS/FAIL testing.

NOTE: *Raw capture never marks connections as wrong — it is a pure measurement view. The result banner reads “RAW CAPTURE — N connection(s): X signal pair(s), Y group(s)”.*

8. Exporting the Test Report

- Step 1: After a test completes, click the EXPORT button in the toolbar.
- Step 2: The Excel workbook is saved to the reports folder. The filename is:

```
cable_test_<SerialNumber>_<YYYYMMDD_HHMMSS>.xlsx
```

Step 3: The workbook is built from the report template and contains four sheets:

- Test Report — header info, firmware version, PASS/FAIL badge (with detected orientation), and the full pin connection table.
- Connection Map — compact pin-pair listing with a status column.
- Test History — cumulative record of tests, appended on each export.
- Pin Map — the connector pin reference sheet.

8.1 Report Layout

Section	Contents
Header block (rows 1–12)	Cable info, operator, firmware version, date/time, opens/shorts/mis-wires found, test duration.
Result badge (row 13)	Large PASS/FAIL banner with signal count, GND connection count, and detected orientation (A→B or B→A).
Column headers (row 16)	# | J1 Pin | Pass Direction | J2 Pin
Row 1 — GND group	Merged cell listing all J1 GND pins and J2 GND pins, grouped by connector row.
Rows 2+ — Signal pairs	One row per signal pair: J1 pin, ↔, J2 pin with ✓ / ✗ / MISSING status.
Footer row	Generation timestamp and software name.

NOTE: *All report tables use complete "All Borders" grids. The title bar, coloured PASS/FAIL banner, and footer are intentionally left without borders.*

9. Connector Pin Layout Reference

Both J1 and J2 connectors carry 116 pins arranged in 7 rows (A–G). The table below summarises the signal/GND assignment pattern.

Row	Pins	Signal Pins (J1)	GND Pins (J1)
A	17	A1, A3, A5, A7, A9, A11, A13, A15, A17 (odd)	A2, A4, A6, A8, A10, A12, A14, A16 (even)
B	17	B2, B4, B6, B8, B10, B12, B14, B16 (even)	B1, B3, B5, B7, B9, B11, B13, B15, B17 (odd)
C	17	C1, C3, C5, C7, C9, C11, C13, C15, C17 (odd)	C2, C4, C6, C8, C10, C12, C14, C16 (even)
D	17	D2, D4, D6, D8, D10, D12, D14, D16 (even)	D1, D3, D5, D7, D9, D11, D13, D15, D17 (odd)
E	16	E2, E4, E6, E8, E10, E12, E14, E16 (even)	E1, E3, E5, E7, E9, E11, E13, E15 (odd)
F	16	F1 - F16 (all signal)	-
G	16	G1, G3, G5, G7, G9, G11, G13, G15 (odd)	G2, G4, G6, G8, G10, G12, G14, G16 (even)

9.1 Cable Signal Mapping (Default cable, A→B)

J1 Row (Signal Pins)	Connected J2 Row
A (odd: A1, A3…A17)	D — same pin number (A1 → D1)
B (even: B2, B4…B16)	C — same pin number (B2 → C2)
C (odd: C1, C3…C17)	B — same pin number (C1 → B1)
D (even: D2, D4…D16)	A — same pin number (D2 → A2)
E (even: E2, E4…E16)	G — same pin number (E2 → G2)

F (all: F1…F16)	F — straight-through (F1 → F1)
G (odd: G1, G3…G15)	E — same pin number (G1 → E1)

NOTE: *When the cable is inserted reversed (B→A), the mapping is mirrored (e.g., J1.D1 → J2.A1) and the software still reports PASS — see Section 5.4.*

10. Troubleshooting

Symptom	Likely Cause / Action
COM port not listed	USB driver not installed or cable unplugged. Install FTDI VCP driver, replug USB.
'Connection failed' on Connect	Wrong COM port selected. Check Device Manager for the correct port number.
Every row shows '✗ expected …'	Cable inserted reversed AND an old software build. Update to the latest software — it auto-detects B→A orientation and passes.
Different / unknown cable shows all FAIL	You are testing against the wrong profile. Select "Raw capture (no reference)", run, then Learn it as a profile.
Specific signal pair MISSING	Open circuit on that wire. Inspect cable and connectors for a broken pin.
Specific signal pair MISWIRE	Wrong J2 pin detected. Check cable wiring against the mapping table.
Test takes > 15 seconds	USB latency. Close other applications using the COM port.
Excel export fails / no template	openpyxl not installed (pip install openpyxl); keep cable_test_report_template.xlsx next to the app, or rebuild the EXE (the template is bundled).
POWER LED off	12 V supply not connected or fuse blown. Check power connector polarity.

11. About

Product	Cable Jig Tester — Cable Continuity Test System
Software	NEU_CABLE_JIG_TEST (PC app updated 2026-07-18)
Firmware	v4.10
Developed by	NEU Corporation
Contact	Jasonquay@corporation.com
Document Rev.	1.1
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